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# Phylum Apicomplexa: Sporozoan or Coccidian

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*AAU, CHS, DMIP, 2023*



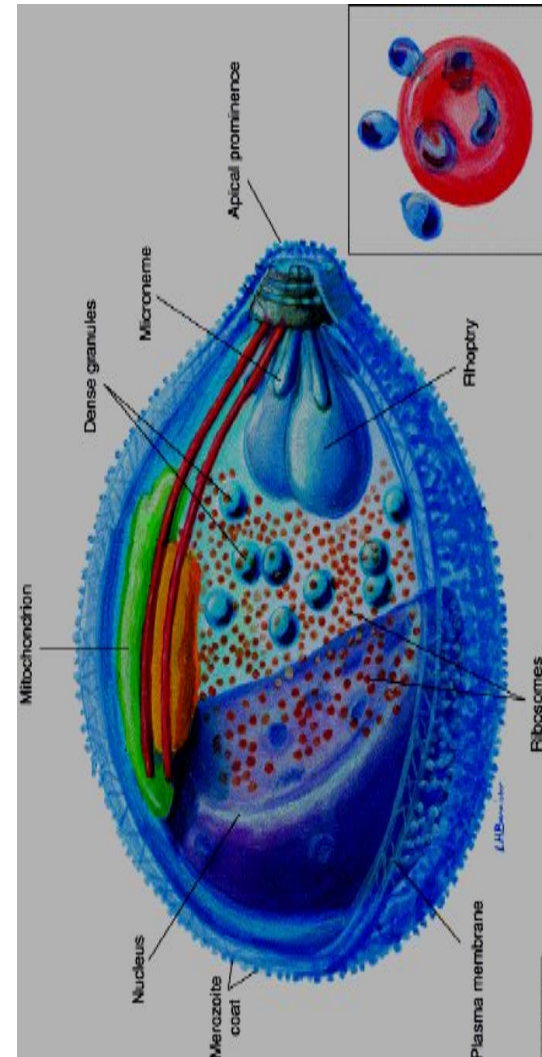
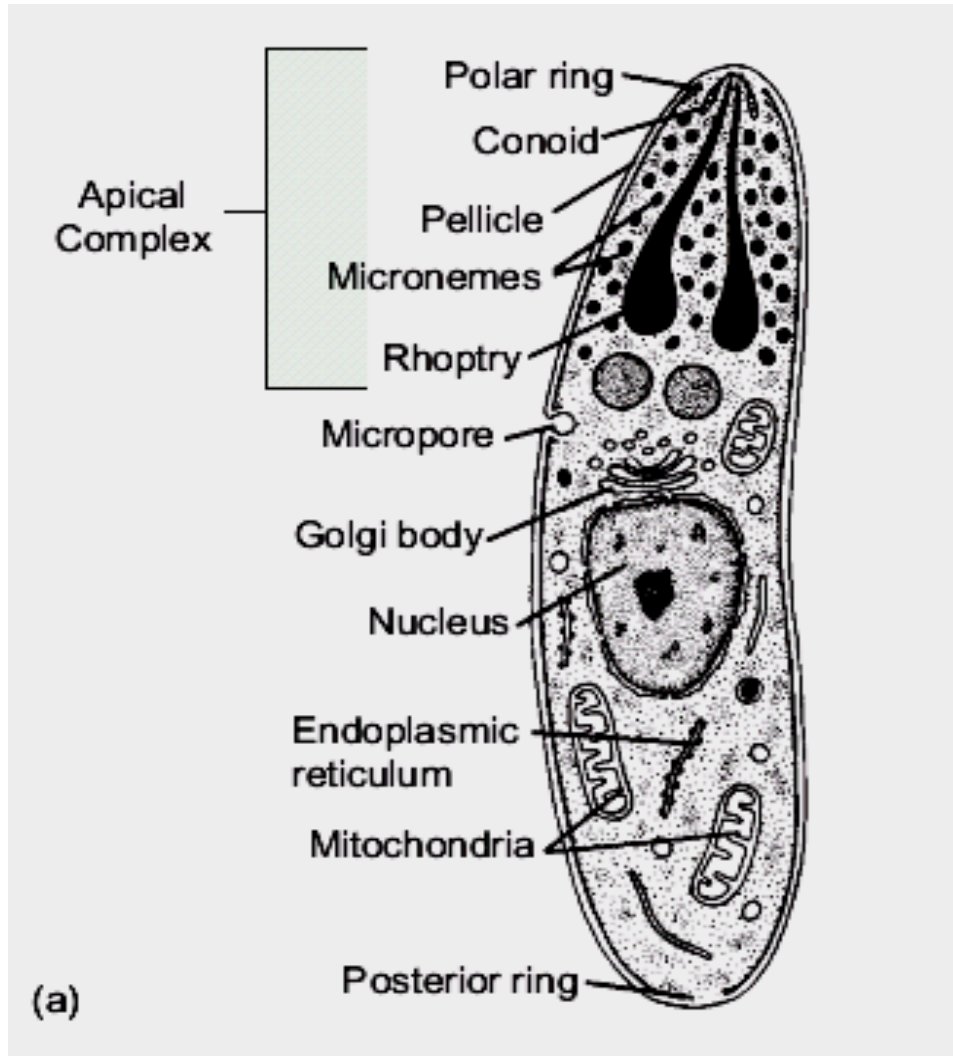
# Phylum Apicomplexa (sporozoa)

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## Class Coccidia

- **Blood and tissue coccidia**
  - **Plasmodium**
  - **Toxoplasma**
  - Sarcocystis
  - Haematozoa
  - Babesia
- **Intestinal Coccidia**
  - **Isospora**
  - **Cyclospora**
  - **Cryptosporidium**

# APICOMPLEXA



# Malaria – Plasmodium species

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- Malaria is a disease caused by members of the genus *Plasmodium* in the phylum **apicomplexa**
- Currently five distinct species infect humans.
  - *P. falciparum*, *P. vivax*, *P. ovale*, *P. malariae* & *new species*  
*P. knowlesi*
- Each species differ in regards to their **morphology, details of their life cycles, and their clinical manifestations**
- *Plasmodium* spps are generally **host specific** & **vector specific**

# General feature of *Plasmodium* species

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- Obligate intracellular parasites (liver cell & RBC)
- **Life cycle**
  - Heterogeneous life cycle: Alternation of generation ~ alternation of hosts, Sexual and asexual reproduction
  - Requires two hosts:
    - Vertebrate hosts: reptiles, birds, rodents, monkeys and human (IH)
    - Arthropod vector: female anopheles mosquitoes (DH)

# Epidemiology of malaria

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## ❖ Wide spread species:

- *P. falciparum*: most prevalent in the hotter and more humid regions of the world
- *P. vivax*: more common in temperate region than in tropics

## ❖ Less widespread species

- *P. malariae*: confined mainly to tropical Africa (25%)
- *P. ovale*: low & restricted distribution occurs primarily in tropical west Africa (10%)
- A new plasmodium species, *P. Knowlesi* in south east asia

# Burden in Ethiopia

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- *P. falciparum* = 60%
- *P. vivax* = nearly 40%
- *P. malariae* = 1% cases, focal distribution like in Humera
- *P. ovale* = less than 1% cases, found in Setit Humera, Gambela & Arbaminch
- The risk of malaria varies highly from season to season and from place to place
- Seasonal transmission mainly depends on rain fall and temperature

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- **Daga zone** (> 2,500masl) mean annual temperature of **10-15°C**, is **malaria free (theoretical)**
  - **Woyna daga zone** (1,500-2,500m asl) mean annual temperatures range from **15-20 °C**
  - Malaria most often occurs below 2,000 m asl, with short-lived transmission following the rains
  - **kolla zone** (< 1,500masl), mean annual temperatures are 20-25 °C, malaria transmission is **endemic**

# Endemicity of malaria

| <b>Endemicity of malaria</b> | <b>Parasite rates<sup>a</sup></b> | <b>Spleen rate<sup>b</sup></b> | <b>Indicates</b>                                            |
|------------------------------|-----------------------------------|--------------------------------|-------------------------------------------------------------|
| Hypoendemic                  | <10%                              | <10%                           | Transmission is low and malaria is not an important problem |
| Mesoendemic                  | 11–50%                            | 11–50%                         | Transmission is moderate                                    |
| Hyperendemic                 | 51–75%                            | 51–75%                         | Transmission is intense but seasonal (rainy season)         |
| Holoendemic                  | >75%                              | >75%                           | Transmission is intense and constantly present.             |

· <sup>a</sup>Parasite rate- % of children between 2–10 years showing malaria parasites in their blood films.

· <sup>b</sup>Spleen rate- % of children between 2–10 years showing enlargement of spleen.

## ➤ Two malaria transmission periods in Ethiopia :

**Major : September to December** after main rainy season

**Minor: April-June** following small showers of rain in autumn

## Characteristics of stable & unstable malaria:

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|              | <i>Stable malaria</i>                                          | <i>Unstable malaria</i>                            |
|--------------|----------------------------------------------------------------|----------------------------------------------------|
| Transmission | Uniform, throughout the year                                   | Seasonal transmission                              |
| Immunity     | Potent resistance in the community due to intense transmission | Lack of community immunity due to low transmission |
| Age          | Children                                                       | All age group                                      |
| Control      | Difficult                                                      | Easier                                             |

- Malaria transmission is either stable or unstable in Ethiopia
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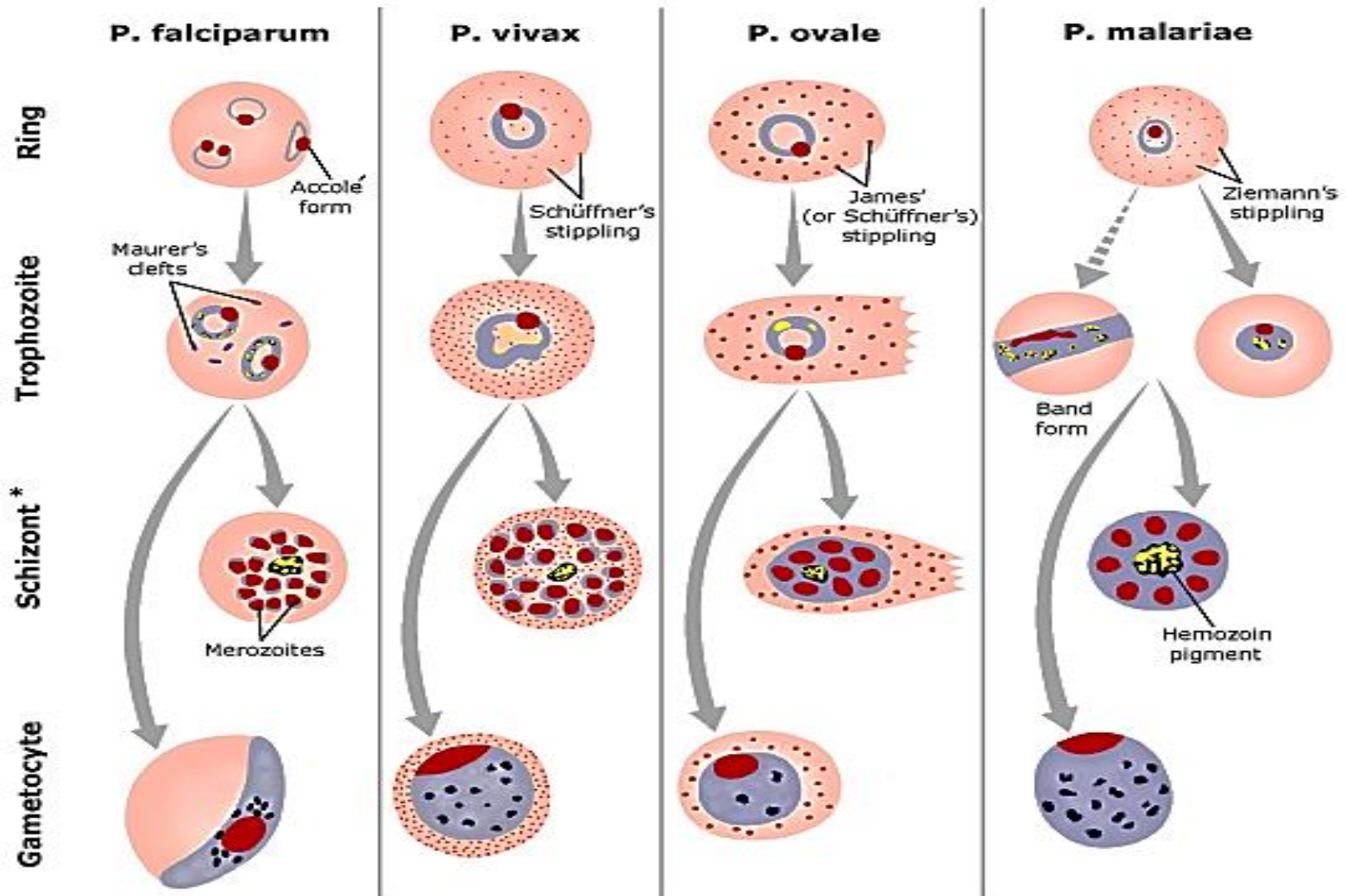


# Developmental stages of malaria

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- ▶ **Sporozoite:** develops in the mosquito salivary gland
- ▶ **Hepatic schizont:** actively dividing, multinucleated form in hepatocytes
- ▶ **Trophozoite:** metabolically active form living within the RBC
- ▶ **Erythrocytic schizont:** multinucleated stage in a RBC resulting from asexual multiplication of trophozoite.
- ▶ **Merozoite:** infective schizont components escape from hepatocyte or RBC
- ▶ **Gametocyte:** morphologically distinctive sexual (male or female) form which develops from some trophozoites in RBCs

# Morphological stages...cont



# Transmission and life cycle of Malaria

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## Principal mode of transmission:

- Bites of infected female anopheles mosquitoes injecting **sporozoite**
- **pierces the skin easily** and **injects saliva** to stop the blood clotting whilst taking meal (anticoagulant + anaesthetic)

### *Female Anopheles Mosquito*



- Mosquito transmission depends:

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- ▶ **susceptibility of anopheles species:** >200 known species of *Anopheles*, 60 of them are considered to be vectors of malaria
- ▶ **Feeding habits:** whether zoophilic or anthropophilic
- ▶ **Density:** distribution of the vector
- ▶ **Longevity:** the life span of the mosquitoes
- ▶ **Climatic factors:** temperature, humidity, rainfall, wind..

- In Ethiopia : *A. gambiae*, *A. funestus*, *A. pharonsensis*, *A. arebiansis* & *A. nili* are main vectors. most epidemic is due to ***A. arabiensis***

# Other mode of transmission

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## Blood transfusion (Transfusion malaria):

- This is fairly common in endemic areas, donor may remain infective for **years**
- Infective stage: **trophozoites/merozoite**
- shorter incubation period, because no **exo-erythrocytic schizogony**, **No possibility of relapse** (*P.vivax*/*P.ovale*)
- clinical features & management of cases are the same as naturally acquired infection



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- ***Mother to the growing fetus (congenital malaria)***
    - Occurs in 5 % of new borne whose mothers are infected
    - Relatively rare although **placenta is heavily infected**
    - Congenital malaria is more common in first pregnancy, among non - immune populations
  - ***Needle stick injury:***
    - Accidental transmission can occur among **drug addicts** who share syringes and needles

# Life cycle

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## ❑ Require two host

### ■ Human:

- Intermediate host
- Asexual reproduction
  - Liver cell, RBC

### ▶ Human cycle

- ▶ Exo-erythrocytic schizogony
- ▶ Erythrocytic schizogony & gametocytogenesis in RBC

### ■ Mosquitoes:

- Definite host
- Sexual, asexual reproduction

### ▶ Mosquitoes cycle

- Sporogony
- Gametogony

# Life cycle in the mosquito

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- Female *Anopheles* mosquito picks up mature **gametocytes**
- ▶ In the mosquito's mid gut, a microgamete(male) penetrates a macro gamete(female) and form a **zygote**
- ▶ The zygotes in turn become motile & elongated form **ookinetes** **which** invade the midgut wall of the mosquito become **oocysts**
- ▶ The oocysts expanding by **asexual multiplication**, **grow**, **rupture**, & release motile **sporozoites**, move to mosquito's *salivary glands*
- ▶ Inoculation of the sporozoites into a new human host

# Life cycle in human

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- ▶ Fraction sporozoites escape destruction are carried rapidly via the blood stream & invade liver hepatic cells
- ▶ Begin their initial asexual replication : *Exo- erythrocytic/ Intrahepatic / Pre-erythrocytic schizogony*
- Within **5-15 days mature** into: ***pre-erythrocyte (PE) Schizonts*** containing 10,000-30,000 merozoites
- Rupture the swollen liver cells & release merozoites
- ***P. falciparum & P. malariae*** : mature and released simultaneously from liver, no relapse

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- ▶ *P. vivax* & *P. ovale*: *parasites* may remain latent in the liver (hypnozoite) and cause relapse infection
  - A proportion of the merozoites are phagocytosed & destroyed
  - The remaining enter in to red cells starts erythrocytic schizogony
  - Intracellular merozoites develop in to trophozoites ('ring form')
  - When the trophozoites fully developed, schizogony takes place and forms schizont containing merozoites

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- Development to erythrocytic schizont in *P. falciparum* takes place in the **capillaries of deep tissue**
  - **The mature schizont rupture from red blood cells releasing merozoites, malaria pigment and toxins in to plasma**
  - Merozoites , which are not destroyed by host immune system infect **new red blood cells,**
  - **Initiates further cycle of erythrocytic schizogony with more red blood cells begin destroyed.**

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- After several erythrocytic schizogony cycle , some of the trophozoites in the RBC develop to male/female gametocytes
  - *P. vivax* , *P. ovale* and *P. malariae* at least two cycle : erythrocytic schizogony
  - *P. falciparum*, asexual parasites stay in the circulation for 10 days
  - The gametocytes are now ready to be ingested by an *Anopheles* mosquito during a blood meal

**i** = Infective Stage  
**d** = Diagnostic Stage

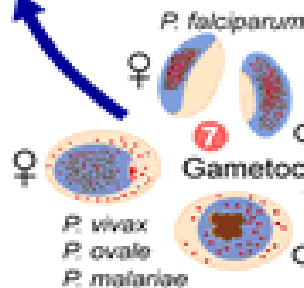
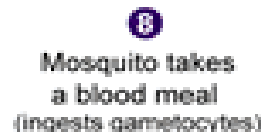
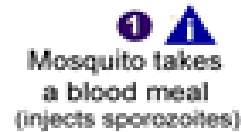
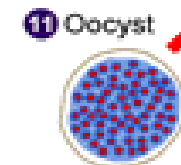


<http://www.dpd.cdc.gov/dpdx>

### Mosquito Stages



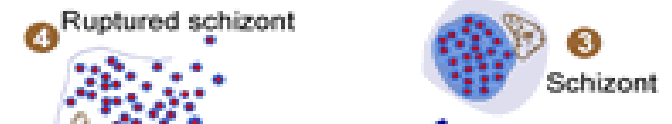
### Sporogonic Cycle



### Human Liver Stages



### Exo-erythrocytic Cycle



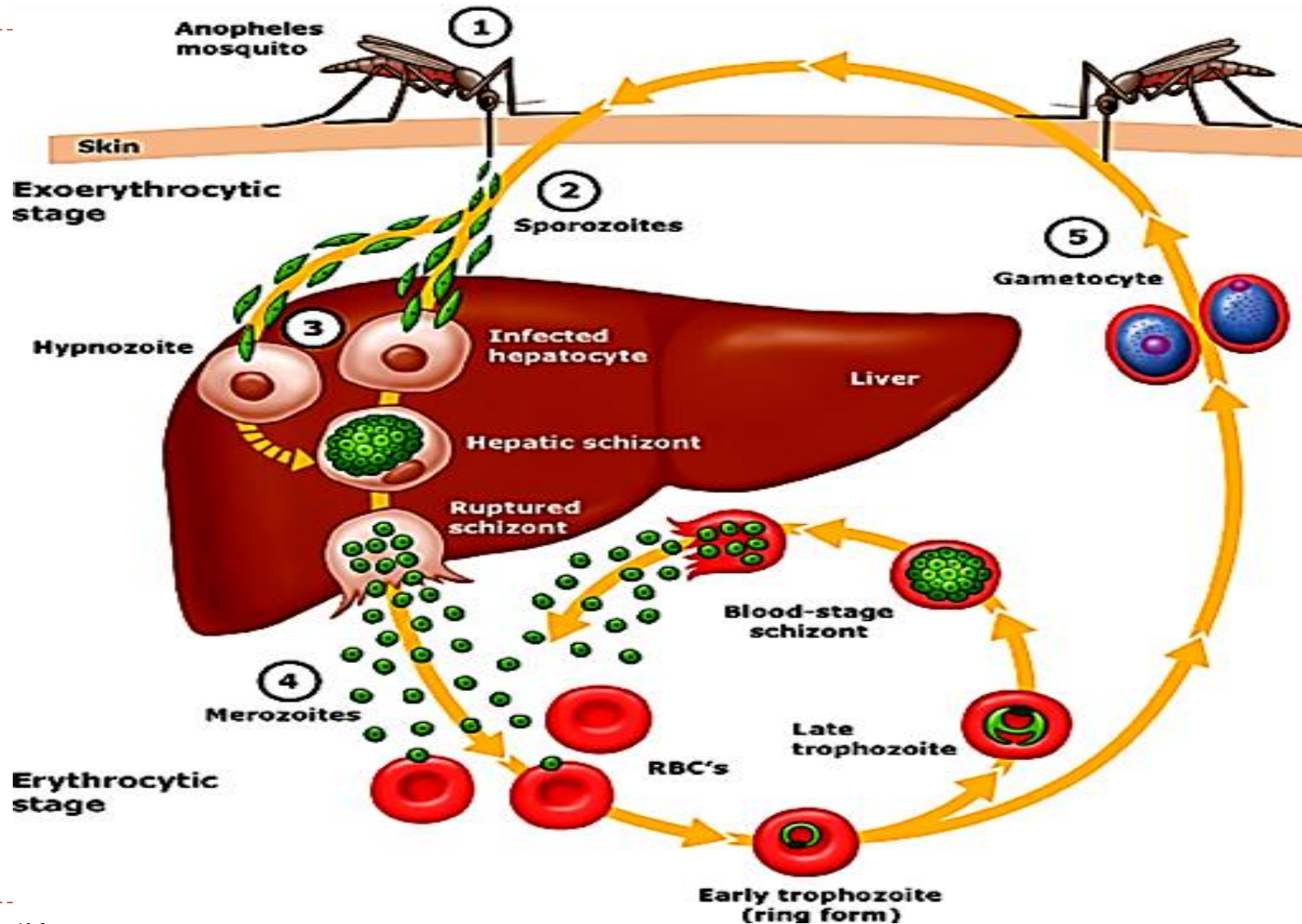
### Human Blood Stages



### Erythrocytic Cycle



## Life cycle and transmission...cont



## Comparison of malarial parasites

|                           | <i>Pf</i>   | <i>Pv</i>     | <i>Po</i>     | <i>Pm</i>    |
|---------------------------|-------------|---------------|---------------|--------------|
| Tissue schizogony         | 8 – 25 days | 8 - 27days    | 9 - 17days    | 15 - 30 days |
| Erythrocytic phase        | 48 hours    | 48 hours      | 48 hours      | 72 hours     |
| Red cells affected        | All         | Reticulocytes | Reticulocytes | Mature RBC's |
| Merozoites per schizont   | 8 - 32      | 12 - 24       | 4 - 16        | 6 - 12       |
| Relapse from Hypanozoites | No          | Yes           | Yes           | No           |



# Clinical Features & pathology

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➤ Pathology in malaria is manifested in two basic forms

➤ Inflammatory reaction and Anemia

✚ Characterized by **acute febrile attacks**:

✚ caused by the release of toxins (when erythrocytic Schizonts rupture) stimulate secretion of cytokines

✚ Manifestations & severity depend on **parasite spps, parasitemia & host status** (immunity, general health, nutritional state, genetics)

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- ▶ *P.vivax*, *P.ovale*, *P. malaria* = **benign**, *P. falciparum* = **malignant**
  - ▶ In *P. falciparum* massive lysis of erythrocytes occurs, which may result in dark-colored urine (black water fever; intravascular hemolysis, hemoglobinuria, kidney failure)
    - ▶ abnormally high levels of hemoglobin in urine and blood.
    - ▶ Fever, vomiting with blood, and jaundice
  - ▶ **Recrudescence**: Due to the low level parasites in blood as a result of inadequate treatment (*P. malariae* & *P.falciparum*)

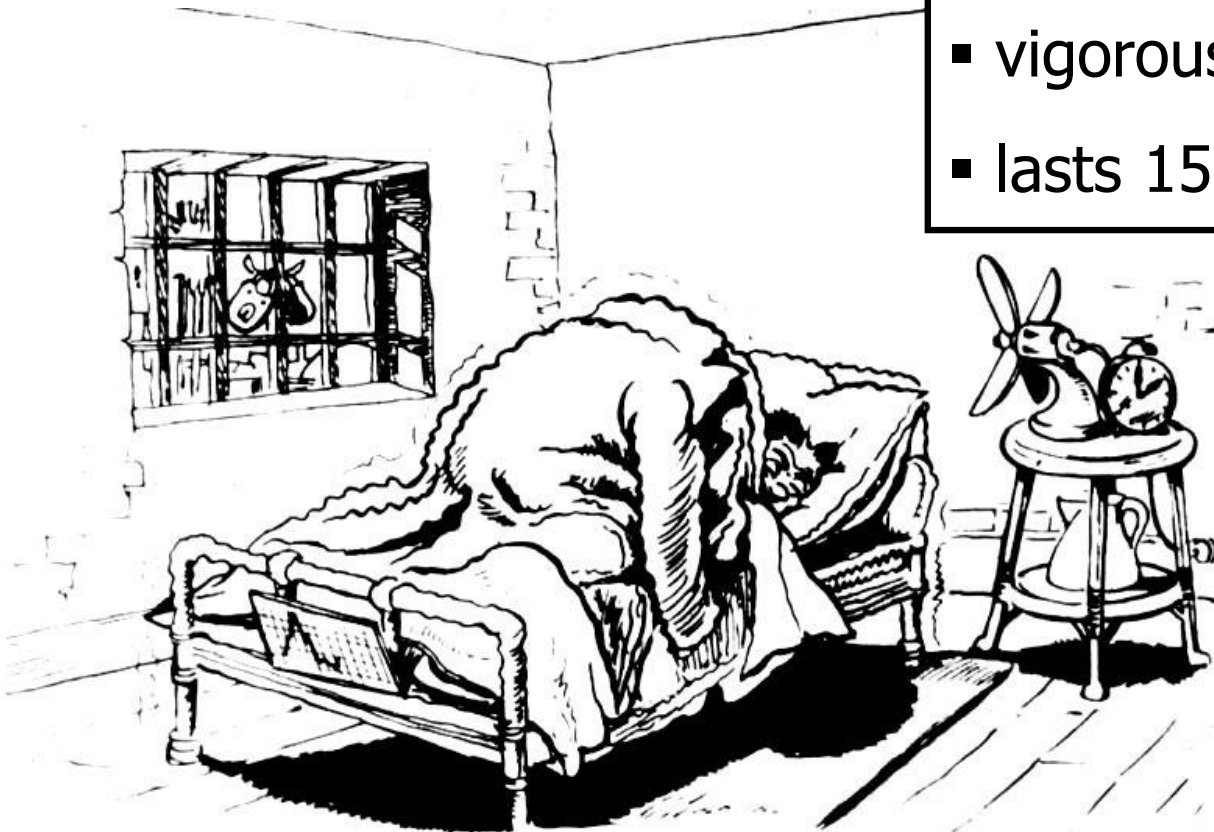
# Clinical Features: Prodromal or onset of Symptoms

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- Prodromal period precede febrile attack (malaria paroxysm) by 2-3 days with a flu like symptoms ranging from mild to severe:
  - Malaise, fatigue, headache, muscle pain, nausea, anorexia
- Malaria paroxysms has three successive stage (4-8 hr):
  - **cold stage, hot stage and sweating stage**
  - Periodic febrile episodes after initially irregular fever
  - Accompanied by splenomegaly, hepatomegaly (slight jaundice), anemia

## **cold stage**

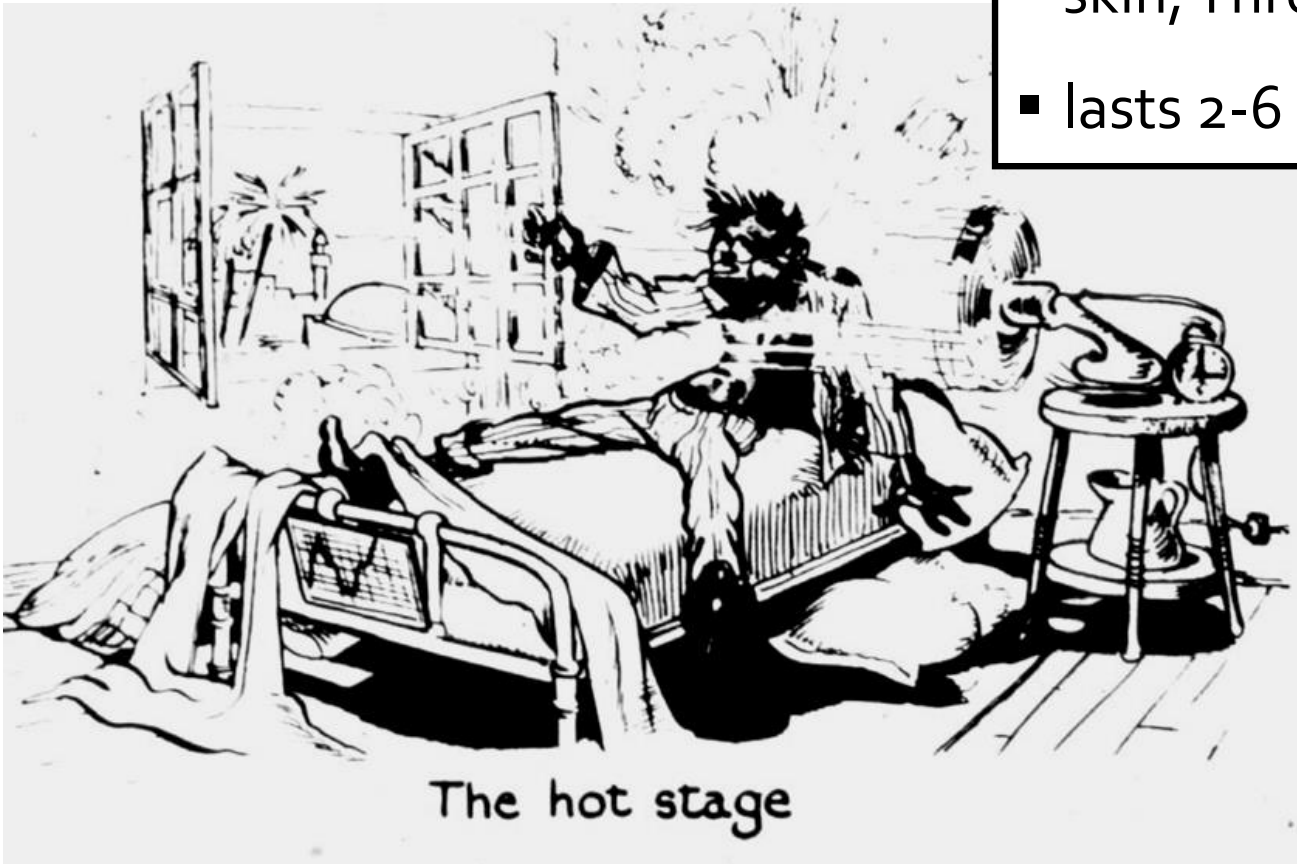
- Feeling of intense cold
- vigorous shivering, rigor
- lasts 15-60 min



**The cold stage**

## Hot Stage

- intense heat, Dry burning skin, Throbbing headache
- lasts 2-6 hours



## sweating stage

- profuse sweating
- declining temperature
- exhausted, weak → sleep
- lasts 2-4 hours



The sweating stage

# Clinical presentation and Pathogenicity

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## Factors for Malignance of *P. falciparum*

- Rapid multiplication, Infected red blood cells become "stick"
  - Infects all age group of red blood cells
  - A single red blood cell can be infected by more than one parasites
  - Erythrocytic schizogonic reproduction takes place in the deep capillaries of organs such as:
    - brain, lung, heart, spleen, bone-marrow, placenta, intestine,
- This makes the diagnosis difficult

# CEREBRAL MALARIA

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- Severe complication of *falciparum* malaria
  - Mortality of 30-50%
  - Associated with *sequestration* in micro-vasculature of brain
    - ▶ Cyto-adherence, Cerebral ischemia, hypoxia, metabolic effects, coma, death

## Predisposing factors for complications of *P. f* malaria

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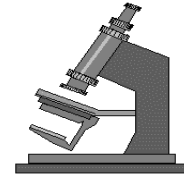
- ▶ **Extremes of age**: children and old ages
- ▶ **Pregnancy**: especially in primigravidae & in 2nd half of pregnancy
- ▶ **Immunosuppressed**: patients on steroids, anti-cancer drugs, immunosuppressant drugs
- ▶ **Splenectomy**: parasitized RBC cleared & intact RBC return to circulation
- ▶ **Lack of previous exposure to malaria**: **non immune**
- ▶ **Pre-existing organ failure**: aggravate the pathogenesis

## Genetic factors that provide protection against Malaria

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- Nature of hemoglobine
  - Hgb S (Sickle cell anemia trait): *P. falciparum*
- Enzyme content of erythrocyte
  - Glucose-6-phosphate dehydrogenase deficiency: *P.f*
- Ovalocytosis: *P.f* & *P.v*
- Duffy blood group antigens (Fya and Fyb) negative RBCs: *P.v*

# Malaria Diagnostics Approaches



**Clinical Diagnosis**

**Malaria Diagnosis**

**Laboratory diagnosis**

**Microscopic**

- Thin film
- Thick film

- Immunological Ag /enzyme
  - ❖ RDT
    - ICT Malaria Pf
    - ParaSight F
    - OptiMAL
  - ❖ Ab-ELISA

**Molecular  
PCR**

# Clinical diagnosis

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- ▶ Based on **clinical signs and symptom**:
- ▶ **Fever**, Chills, perspiration, anorexia, headaches, vomiting, and malaise
- ▶ It is **inexpensive** to perform and requires no special equipment or supplies.
- ▶ Are **non-specific** and symptoms overlap with those of other febrile illnesses.
- ▶ Diagnosis of malaria based on Clinical grounds alone is therefore **unreliable- over diagnosis**

# Confirmed Diagnosis of Malaria

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- ▶ All clinically suspected malaria cases require **laboratory examination and confirmation**.
- ▶ Only in case where laboratory confirmation is not possible start treatment immediately.
- ▶ Parasitological confirmation is done by **thin-thick blood smear microscopy examination or by dipstick (Rapid Diagnostic Test [RDT])**.

# Laboratory diagnosis Techniques

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## I. Microscopy examination

- Peripheral smear study: blood films (Thick and thin)
- **Concentrating Malaria Parasites** : Concentrating parasite in venous blood by centrifugation
- Buffy Coat preparation / **Quantitative buffy coat( QBC)**

## II. Immunologic/Biochemical techniques

- ▶ Detection of malaria antigen, antibody & parasite products

## III. Molecular diagnosis techniques

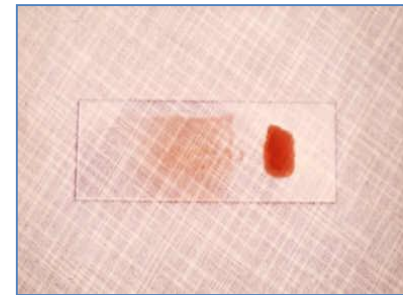
# Microscopic Examination Of Blood Film

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- ▶ Established method for confirmation of malaria:  
the gold standard

## Requirements:

- ▶ Carefully collected blood specimen/sample
- ▶ Well prepared blood films
- ▶ Well stained smears
- ▶ Careful examination of the smears



# Collection of Blood Specimen

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## I. Collect sufficient quantity of blood

1. Capillary blood from finger prick , toes, or ear lobes –best
2. Venous blood.(EDTA anticoagulant)
3. In obstetric practice, cord blood & placental impression smears

## II. Time of collection

- When the patients feels febrile
- Before anti-malaria drugs are given to the patients
- Note:
  - » A negative test DOES NOT rule out malaria.
  - » Repeated tests may have to be done in all doubtful cases



# I. Preparation of thick films

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- ▶ slides must be clean, free from dirt, grease and fingerprints
- ▶ Mix the blood
- ▶ Using micropipette apply 6 $\mu$ l blood on to a microscope slide for thick blood film
- ▶ Spread the blood without excessive stirring to form a smear approximately 1cm<sup>2</sup>, through which newspaper print can be read.
- ▶ This should be approximately 5-10 red blood cells thick
- ▶ Allow to air dry horizontally (without using heat)

## II. Preparation of Thin Films

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- ▶ The microscope slides must be clean as for thick films otherwise the blood will not adhere & smear will be irregular
- ▶ Apply 2  $\mu$ l blood to the end of the slide, place another slide at an angle of 30-45° & bring it towards the drop of blood.
- ▶ As soon as it touches it, the blood will disperse along the width of the slide.
- ▶ Before it reaches the edges, pull the drop along the length of the slide.
- ▶ The correct amount of blood in the drop the film will form a good tail before the end of the slide: The film here should be 1 RBC thick.
- ▶ Air dry, Label slides

## Thick blood film

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- ▶ Good for rapid detection malaria parasites , particularly when they are few
- ▶ About 30 times more sensitive (detecting about 20 parasites/ $\mu$ l)
- ▶ In a thick film the blood is not fixed

## Thin blood film

- ▶ Required to confirm the *plasmodium* species
- ▶ Enabling the parasites to be seen in the red cells
- ▶ Greatly assists in the identification of mixed infection
- ▶ Value in assessing treatment efficacy
- ▶ Gives opportunity to investigate anemia & WBC abnormalities

# Reporting blood films for malaria parasites

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## A. Reporting thick blood films: the plus sign scheme

- 1) 1 – 10 parasites per 100 high power fields .....+
- 2) 11 -100 per 100 high power field.....++
- 3) 1 – 10 in every high power field.....+++
- 4) More than 10 in every high power field ... ++++

## B. Against 200 WBC in thick film is now recommended

- ☐ If no parasites are found after examining **100 fields** (or if **indicated 200 fields**), report the film as:

- Malaria thick film: No parasites seen or no hemo parasite seen

# MALARIA TREATMENT PROTOCOL

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## ❖ Uncomplicated malaria definition:

### **Fever and any of the following:**

- ▶ Headache,
- ▶ Body and joint pains
- ▶ Feeling cold and sometimes shivering
- ▶ Loss of appetite and sometimes abdominal pains
- ▶ Diarrhoea, nausea and vomiting.
- ▶ Splenomegaly

# Differential diagnosis for uncomplicated malaria

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Consider other illnesses, such as:

- ▶ Upper respiratory tract infection (Pharyngitis, tonsillitis, ear infection), pneumonia, measles, dengue, influenza, typhoid fever.
- ❖ Remember that the patient may be suffering from more than one illness.
- ❖ **Uncomplicated malaria treatment (P.F)**
  - ❖ Coartem, Artesunate, Quinine

## Uncomplicated malaria treatment : *P. vivax*

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**Chloroquine** is the drug of choice

- ▶ For the radical treatment of *P. vivax* in addition to chloroquine, **primaquine** is recommended 0.5mg/kg per day for 14 days (primaquine should always be taken with food).
- ▶ Chloroquine can be given to pregnant women and children.
- ▶ Primaquine is not recommended for the children under one year and pregnant women.

## *P. falciparum* and *P. vivax* (mixed infections)

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- ❖ The type of malaria where both infections occurs in patient requires treatment by **Coartem®**.
- If the Pf dipstick is negative and the clinical signs are typical for malaria, treat with Chloroquine (it could be a case of *P. vivax*)
- If the Pf dipstick is negative and the clinical signs don't suggest malaria, do not treat like malaria; look for another illness.
- If there is no possibility of dipstick or slide results, treat the patient based on the clinical signs and symptoms. Treat as if the patient has *P. falciparum*

# Follow-up of uncomplicated malaria:

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## For treatment failure for adult After 14 days:

- ▶ **Quinine** (10mg salt /kg bw three times a day) + **doxycycline** (3.0mg/kg bw once a day) for 7 days.
- ▶ Do not give doxycycline with milk or iron, which will reduce its absorption.
- ▶ If patient is in health facility where microscopy facility is not available patient should be referred to the facility where microscope is available.
- ▶ If refer is not possible treatment should be given Quinine + Doxycycline.

## Follow-up of uncomplicated malaria:

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- Doxycycline should not be given to pregnant or lactating woman, or child aged up to 8 years.

### For pregnant or lactated woman or child less than 8 years:

- ▶ **Quinine** (10mg salt /kg bw three times a day) + **clindamycin** (10mg/kg bw twice a day) for 7days.
- ▶ For small children, (quinine and clindamycin) crush tablets and mix with water and sugar.

# SEVERE MALARIA

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**Severe or complicated malaria definition:**

**Fever and any of the following:**

- ▶ Impaired consciousness
- ▶ Anxiety, palpitation and sweating
- ▶ Convulsions, Fast or difficult breathing
- ▶ Vomiting every feed / unable to feed
- ▶ Pale hands, tongue and inner parts of the eyelid
- ▶ Generalized body weakness
- ▶ Dehydration, Jaundice, Severe malnutrition
- ▶ Dark urine or no urine

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- ▶ A patient who is non responsive should be quickly assessed and managed.
  - ▶ This includes assessment of the airway, breathing and circulation.
  - ▶ The staff at the first level health facility should be able to maintain airway, provide assisted breathing and manage shock if required.
  - ▶ Check blood sugar, if possible!

## Confirmed diagnosis of severe malaria:

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- ▶ All clinically suspected severe malaria cases require laboratory examination and confirmation.
- ▶ Only in case where laboratory confirmation is not possible start treatment immediately.
- ▶ Parasitological confirmation is done by thin-thick blood smear microscopy examination or by dipstick (Rapid Diagnostic Test [RDT]).

# Differential diagnosis for complicated malaria

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- ▶ Consider other illnesses, such as:
- ▶ Measles, meningitis, tonsillitis, dengue, otitis media (ear infection), influenza, pneumonia, typhoid fever, tuberculosis, hypoglycemia.
- ▶ **Severe malaria treatment**
- ▶ Artesunate
- ▶ Quinine injection 20mg/kg

# Prevention and Control

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- ▶ **Avoid mosquito bites** by using impregnated bed nets, Wearing protective clothes, mosquito repellents, Screens
- ▶ **Destroy adult mosquitoes** by: **Indoor residual regular effective spraying**
- ▶ Environmental modification of mosquito breeding sites, Spraying breeding places with effective larvicides, Biological control
- ▶ Treatment of: Active infection & Chemoprophylaxis
- ▶ Health information, donor blood screening before transfusion

# Malaria Vaccine

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## ▶ Mosquirix (RTS, S/ASo1)

- ▶ Endorsed by the World Health Organization In October 2021
- ▶ the first malaria vaccine candidate to address **parasitic infection**
- ▶ is a vaccine that is given to children aged 6 weeks to 17 months to help protect against *P. falciparum*.
- ▶ is a recombinant protein-based malaria vaccine.

